1.5.2 Homogeneous system of linear equations

We recall that a homogeneous system of linear equations is given by

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= 0, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= 0, \\ \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots & \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= 0, \end{aligned} \qquad \ldots (1)$$

where the coefficients $a_{ij}, i = 1, 2, \cdots, m; j = 1, 2, \cdots, n$ are constants. The above system is always satisfied by $x_1 = 0, x_2 = 0, \cdots, x_n = 0$. This solution is called the **trivial solution** of (1). In other words, the system (1) always possesses a solution.

The above system (1) can be put in the matrix form $AX = O_{m \times 1}$, where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, O_{m \times 1} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

We will denote $O_{m \times 1}$ simply by the capital letter O. Since O is the zero column matrix, it is always true that $\rho(A) = \rho([A \mid O]) \leq m$. **So, by Rouché-Capelli Theorem, any system of homogeneous linear equations is always consistent.**

Suppose that $m < n$, then there are more number of unknowns than the number of equations. So $\rho(A) = \rho([A \mid O]) < n$. Hence, system (1) possesses a non-trivial solution.

Suppose that $m = n$, then there are equal number of equations and unknowns:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= 0, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= 0, \\ \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots & \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \cdots + a_{nn}x_n &= 0, \end{aligned} \qquad \ldots (2)$$

Two cases arise.

Case (i)

If $\rho(A) = \rho([A \mid O]) = n$, then system (2) has a **unique solution** and it is the **trivial solution.** Since $\rho(A) = n, |A| \neq 0$. So for **trivial solution** $|A| \neq 0$.

Case (ii)

If $\rho(A) = \rho([A \mid O]) < n$, then system (2) has a **non-trivial solution**. Since $\rho(A) < n, |A| = 0$. In other words, **the homogeneous system (2) has a non-trivial solution if and only if the determinant of the coefficient matrix is zero.**

Suppose that $m > n$, then there are more number of equations than the number of unknowns. Reducing the system by elementary transformations, we get $\rho(A) = \rho([A \mid O]) \leq n$.

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